

EXPERIMENT NO.7

Name:Aryan Anil Patankar

Class:D20A

Roll No:38

Batch:B

Aim:Implement the embedding wallet (Metamask) and transaction using Solidity

Theory:

1. What is MetaMask?

MetaMask is a software cryptocurrency wallet used to interact with the Ethereum blockchain and other EVM-compatible networks (like Polygon, BSC, and Avalanche).

It is primarily known as a "**bridge**" because it allows users to access Decentralized Applications (dApps) directly through a regular web browser (like Chrome or Firefox) without having to run a full Ethereum node.

Core Theoretical Functions:

- **Identity Management:** It generates and manages your private keys locally on your device (not on a central server), giving you full ownership of your digital identity.
- **Web3 Injection:** It injects a JavaScript library called **web3.js** or **ethers.js** into the websites you visit. This allows the website to "talk" to the blockchain.
- **Transaction Signing:** When you want to perform an action on a dApp (like buying an NFT), MetaMask "pops up" to let you review and digitally sign the transaction before it is sent to the network.
- **Network Switching:** It allows users to switch between the "Mainnet" (real money) and various "Testnets" (fake money).

2. What is a Testnet?

A **Testnet** (Test Network) is an alternative blockchain that acts as an exact replica of the main blockchain (Mainnet) but is used specifically for **experimenting, testing, and debugging**.

In blockchain theory, deploying code directly to the Mainnet is risky and expensive because every transaction costs real money (Gas fees) and bugs can lead to permanent financial loss. Testnets solve this by providing a "sandbox" environment.

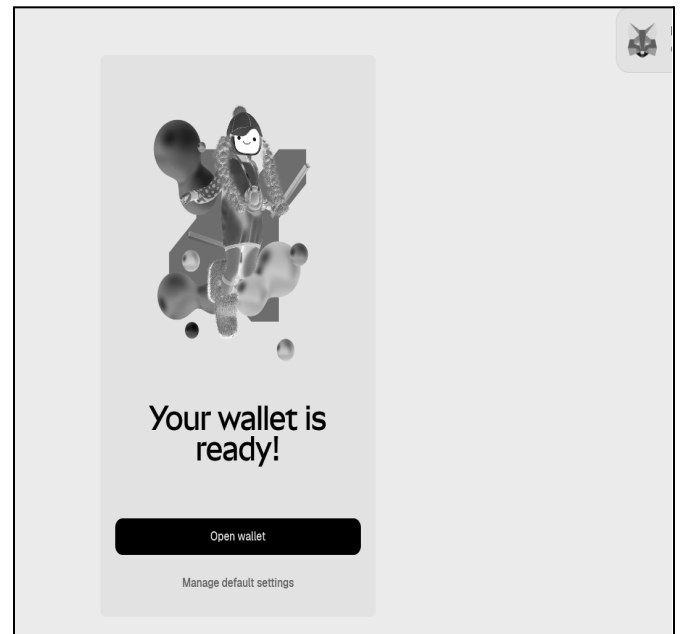
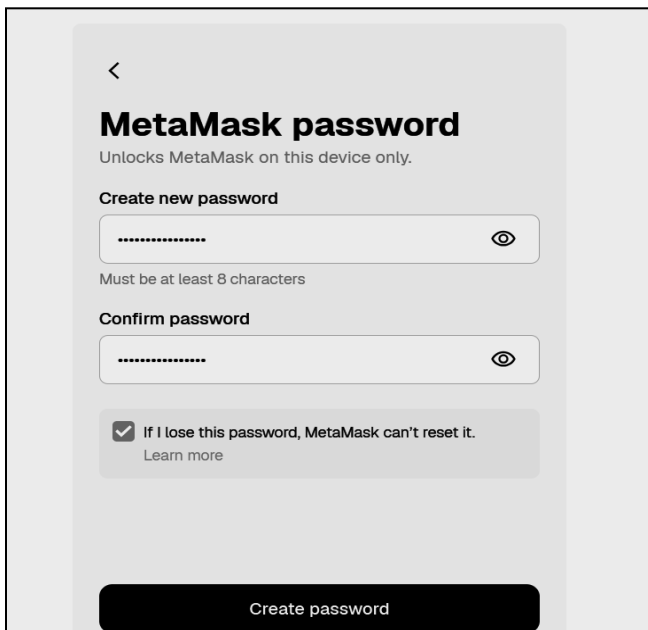
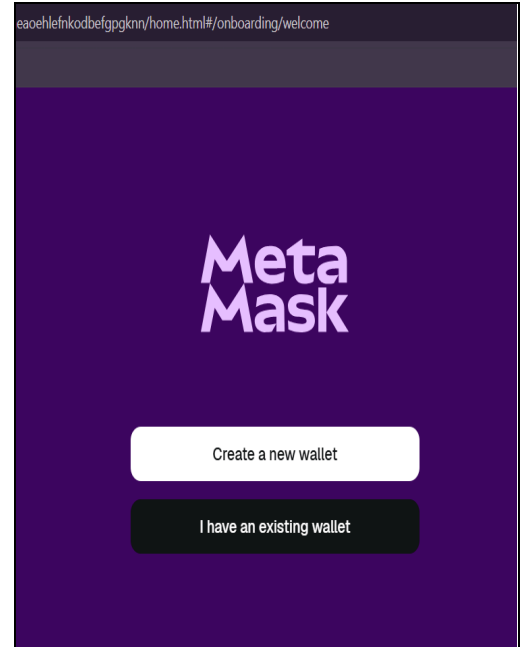
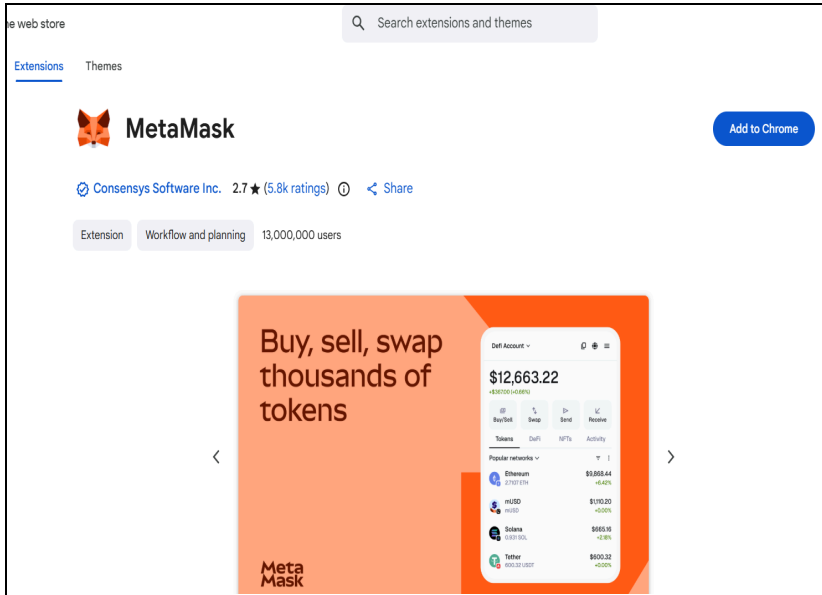
Key Characteristics:

- **No Real Value:** The tokens used on a testnet (e.g., Sepolia ETH) have **zero real-world market value**. You can get them for free from "Faucets."
- **Identical Environment:** It uses the same rules, consensus mechanisms (like Proof of Stake), and smart contract logic as the Mainnet.
- **Safe Playground:** If a developer's smart contract has a bug that crashes the system or "leaks" funds, no real money is lost.
- **Public Access:** Most testnets are public, meaning anyone can join and test their applications before the official launch.

Implementation:

Step 1: Install Metamask

Follow this [link](#) to install the Metamask Google Chrome Extension, and also create the wallet

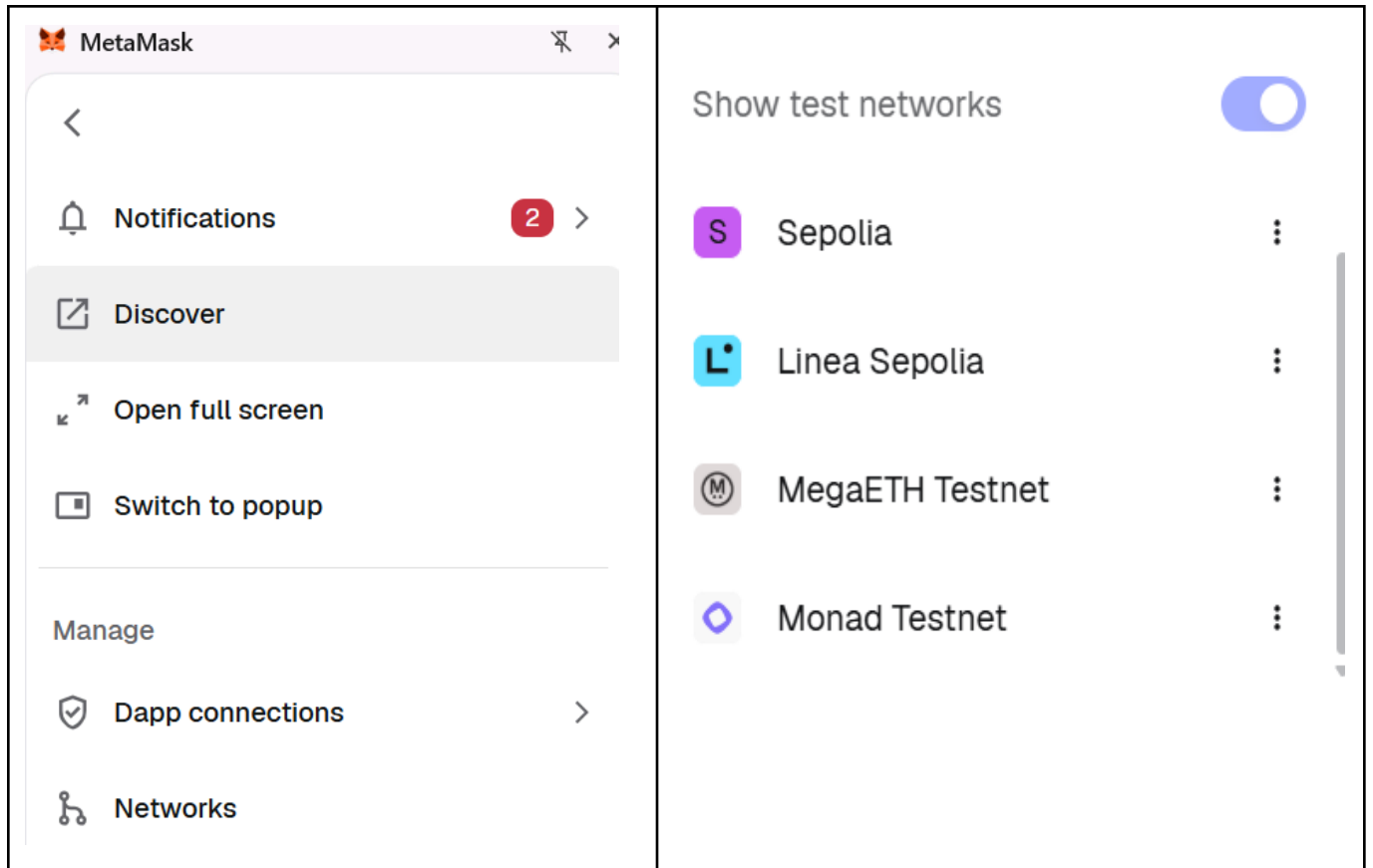


Note :

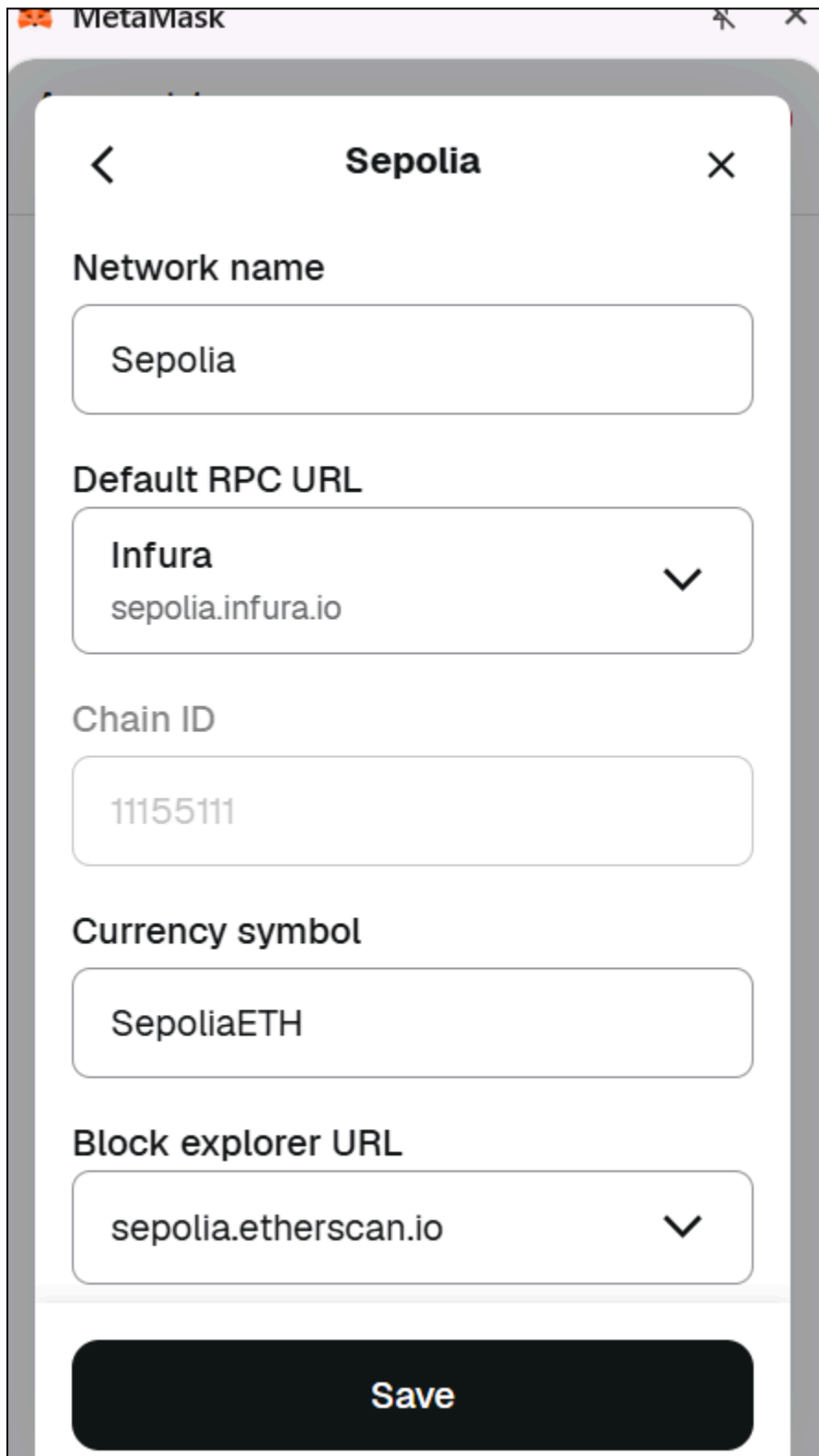
- While creation of the Metamask Wallet, you will be asked to select 12 pass phrase
- Keep the passphrase safely
- It becomes handy to recover your password

Step 2: Enable Test Networks

1. By default, MetaMask shows the "Ethereum Mainnet" (real money).
2. Click the network dropdown (top left) and toggle **Show test networks**.
3. Select **Sepolia**.



Step 3: Configuring the Sepolia Test network



The image shows the MetaMask mobile application interface for adding a new network. The window title is "MetaMask". The screen has a title bar with a back arrow, the word "Sepolia", and a close "X" button. Below the title bar, there are five input fields, each with a label and a rounded rectangular input area. The first field is labeled "Network name" and contains the text "Sepolia". The second field is labeled "Default RPC URL" and contains a dropdown menu with "Infura" and "sepolia.infura.io" below it, and a downward arrow icon. The third field is labeled "Chain ID" and contains the text "11155111". The fourth field is labeled "Currency symbol" and contains the text "SepoliaETH". The fifth field is labeled "Block explorer URL" and contains a dropdown menu with "sepolia.etherscan.io" and a downward arrow icon. At the bottom of the screen is a large black button with the text "Save" in white.

MetaMask

< Sepolia X

Network name

Sepolia

Default RPC URL

Infura
sepolia.infura.io

Chain ID

11155111

Currency symbol

SepoliaETH

Block explorer URL

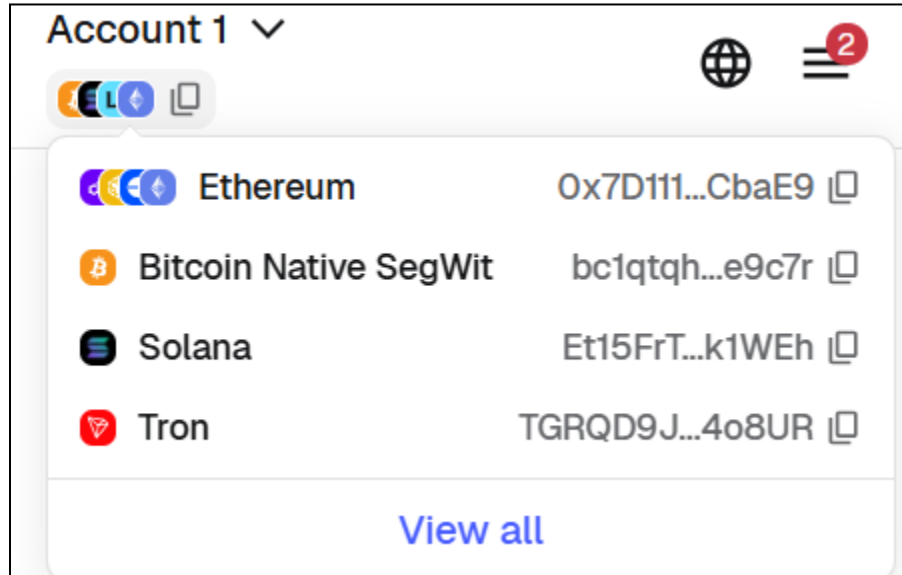
sepolia.etherscan.io

Save

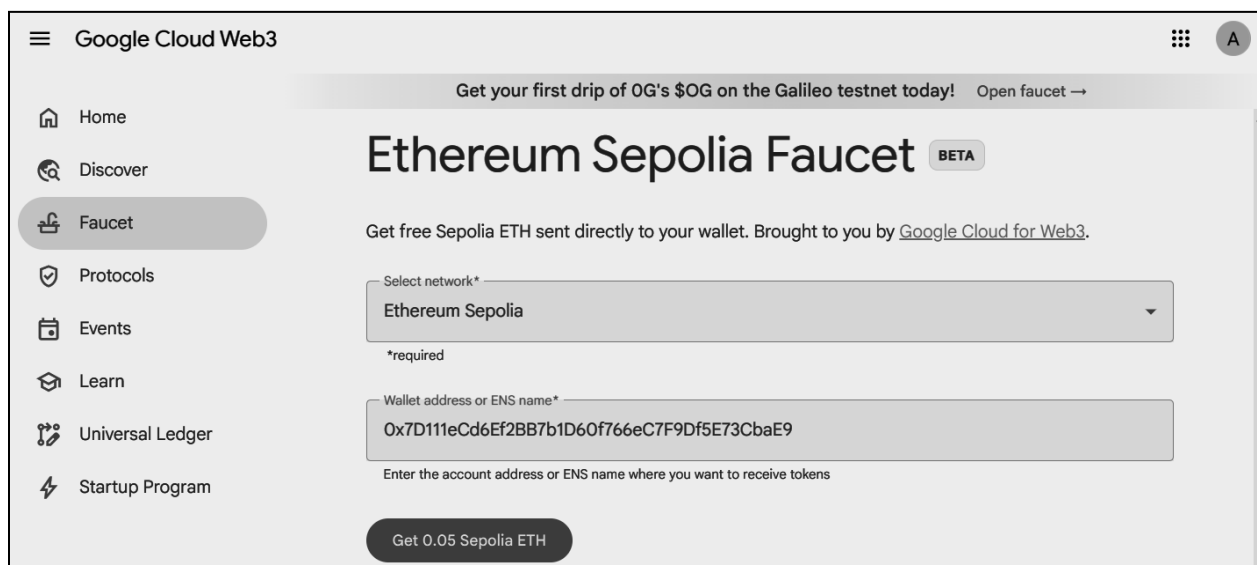
Step 4: Funding the Wallet

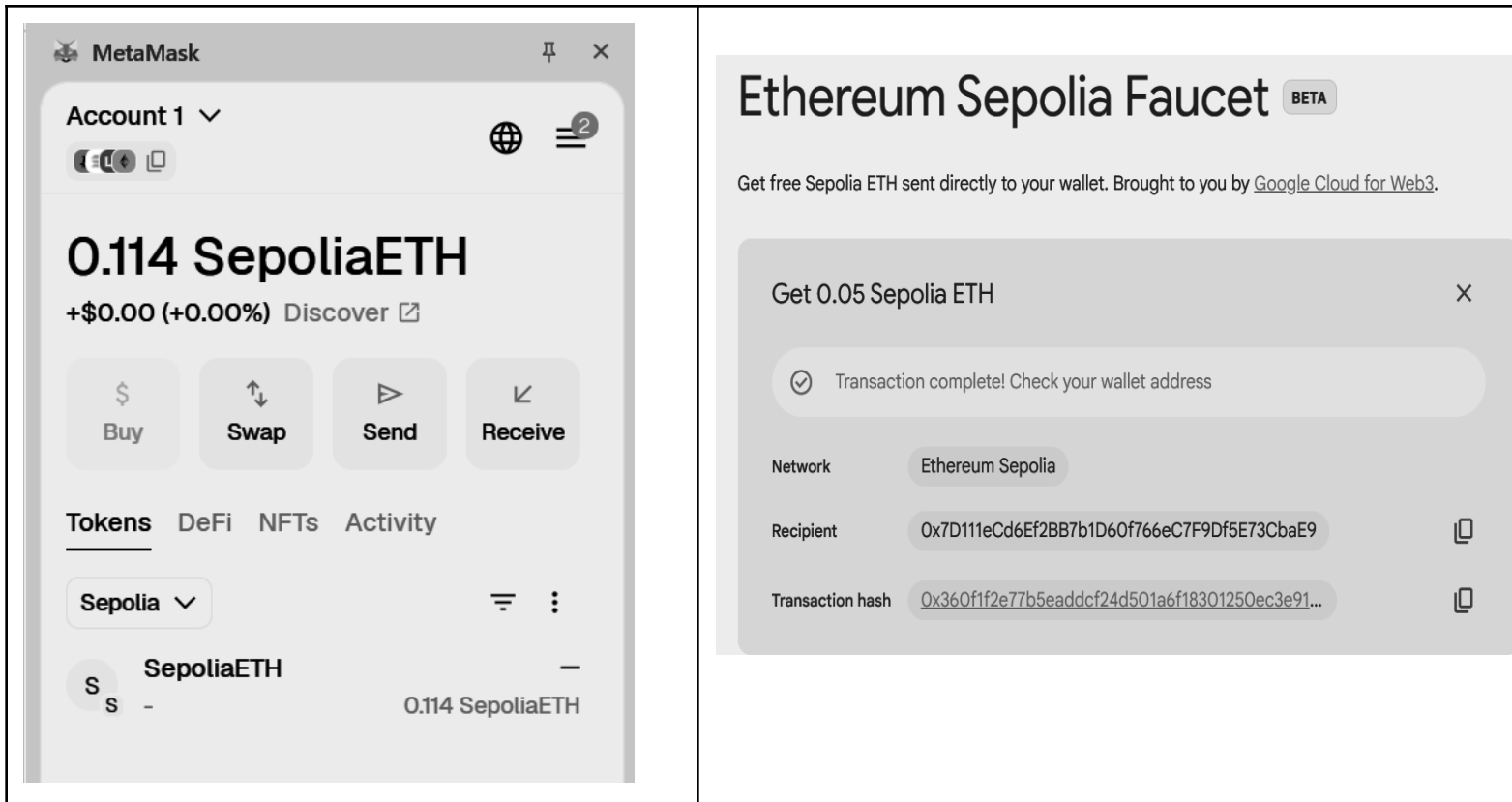
Every action on the blockchain requires a fee called **Gas**. On the Sepolia network, we use "Test ETH" which is free.

1. **Copy Your Address:** Click your account name at the top of MetaMask (it starts with `0x...`) to copy your public address.

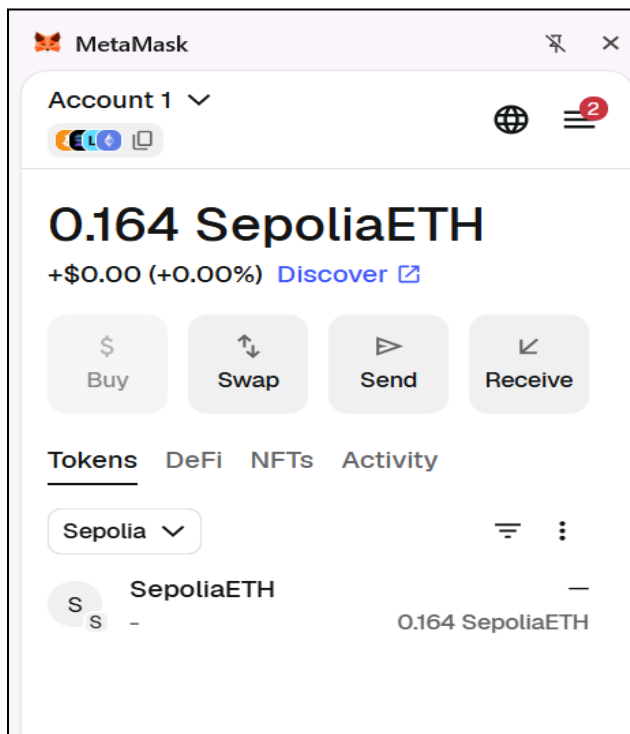


2. **Visit a Faucet:** Use the [Google Cloud Sepolia Faucet](#) to procure SepoliaETH by pasting the wallet address as shown



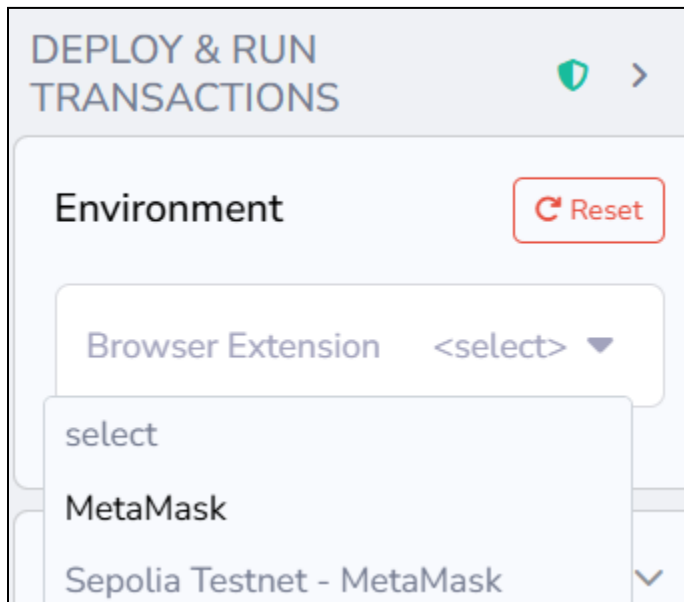


3. **Verify:** Your MetaMask balance should update within 30 seconds.

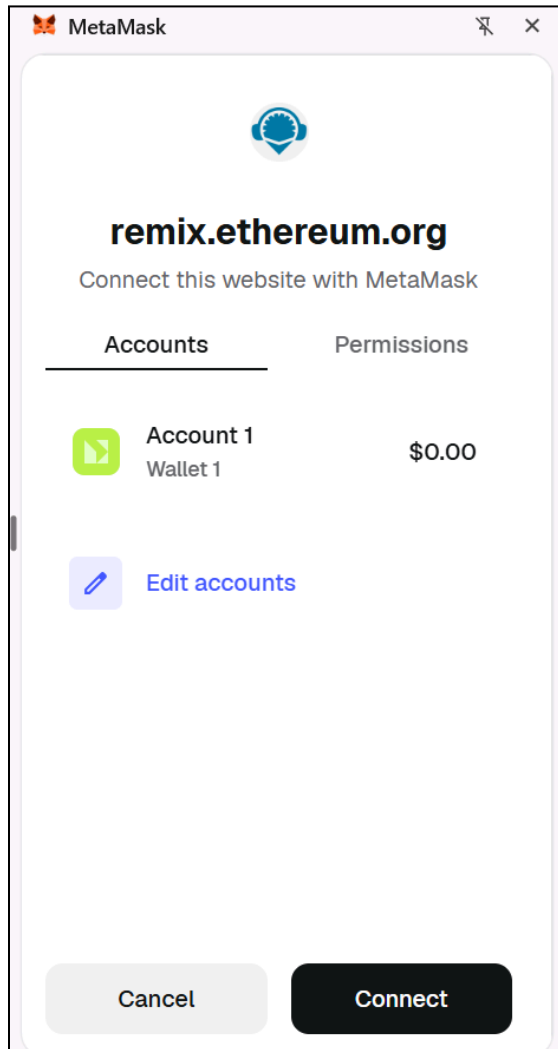


Step 5: Connect MetaMask to Remix IDE

1. Open [Remix IDE](#).
2. Go to the **Deploy & Run Transactions** tab.
3. In the **Environment** dropdown, select **Injected Provider - MetaMask**.

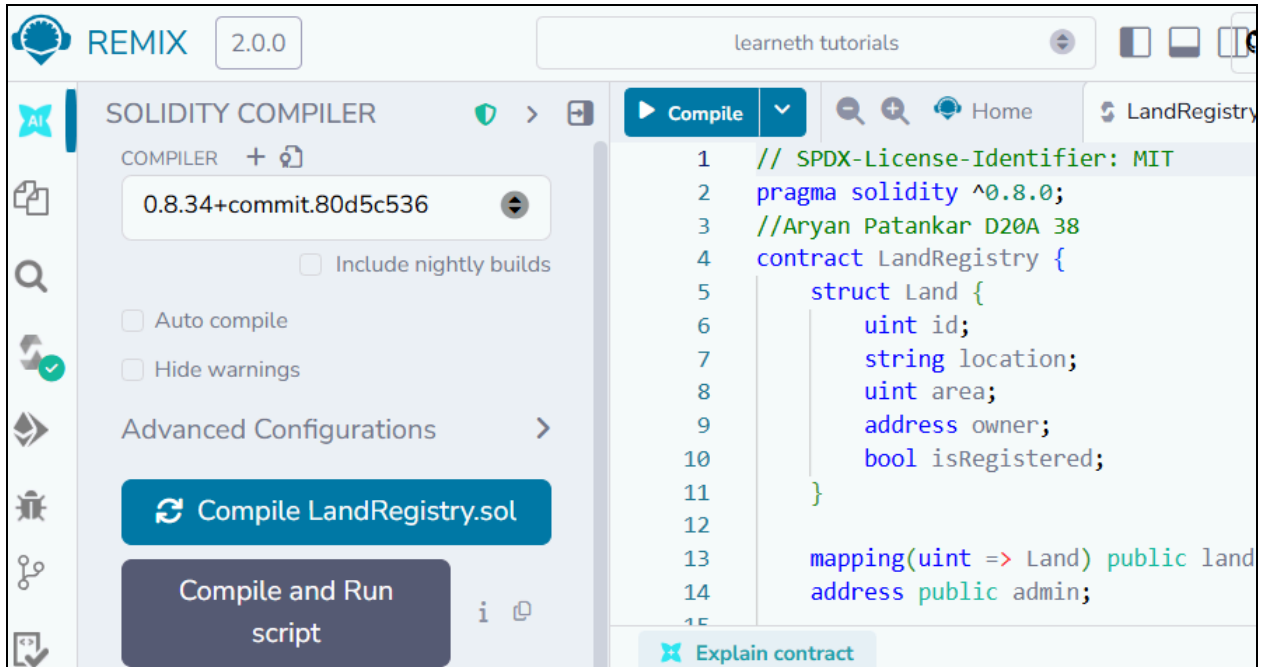


A MetaMask popup will appear; click **Connect** to link your `0x7D11 . . .` account to Remix.



Step 6: Create and Compile the Contract

1. Open your `LandRegistry.sol` file.
2. Go to the **Solidity Compiler** tab.
3. Select Compiler `0.8.0` (or the version in your code).
4. Click **Compile LandRegistry.sol**.



Step 7: Deploy and Verify the transaction

1. Go back to the **Deploy & Run Transactions** tab.
2. Ensure **Injected Provider** is still selected (you should see "Custom (11155111) network").
3. Click the orange **Deploy** button.
4. **MetaMask Action:** A popup will show the gas fee. Click **Confirm**.
5. **Wait:** Look at the bottom terminal. Wait for the green checkmark that says "transacted."

DEPLOY & RUN
TRANSACTIONS

Environment

Reset

Browser Extensi... MetaMask

Account 16 0.163 ETH
0x7d1...cbae9

Deploy Sepolia (11155111) network

LandRegistry
LandRegistry.sol

Compiled

Verify Contract on Explorers

Value 0 wei

Gas limit auto 0

Deploy

MetaMask

Account 1

Deploy a contract

This site wants you to deploy a contract

Estimated changes ? No changes

Network s Sepolia
Request from ? remix.ethereum.org

Network fee ?
0.0002 s SepoliaETH

Speed Market ~12 sec

Cancel

Confirm

After clicking on deploy button

Sepolia Testnet

Search by Address / Txn Hash / Block / Token

Transaction Details

OverviewState

TRANSACTION ACTION

Call

0x60806040

Method by 0x7D111eCd...5E73CbaE9

[This is a Sepolia Testnet transaction only]

Transaction Hash:

0x5ddffc83a10bff63281513255d95206694b867c481de576cc0408c4666d12f4f

Status:

Success

Block:

105279261 Block Confirmation

Timestamp:

10 secs ago (Mar-26-2026 08:18:00 PM +UTC)UTC

From:

0x7D111eCd6Ef2BB7b1D60f766eC7F9Df5E73CbaE9

To:

[0xa96899f7bde961cd19c979285124bb70725ee133 Created]

Value:

0 ETH

Transaction Fee:

0.001242280517391927 ETH

Gas Price:

1.500000021 Gwei (0.000000001500000021 ETH)

Transaction is verified on Sepolia Etherscan



Contract deployment



Status

[View on block explorer](#)**Confirmed**[Copy transaction ID](#)

From

To

 0x7D111...CbaE9

New contract

Transaction

Nonce	0
-------	---

Amount	-0 SepoliaETH
--------	---------------

Gas Limit (Units)	835927
-------------------	--------

Gas Used (Units)	828187
------------------	--------

Base fee (GWEI)	0.0000000021
-----------------	--------------

Priority fee (GWEI)	1.5
---------------------	-----

Total gas fee	0.001242 SepoliaETH
---------------	---------------------

Max fee per gas	0.000000002 SepoliaETH
-----------------	------------------------

Total	0.00124228 SepoliaETH
-------	------------------------------

✓ [block:10527933 txIndex:35] from: 0x713...99CC2
to: 0xe96F6350d2059BA2E357fbDB3fa173eC18c6af16 0xe96...6af16 value: 0 wei
data: 0x786...b844b logs: 2 hash: 0x190...619a1

Debug

^

status	1 Transaction mined and execution completed
transaction hash	0x0e1dd86e32bf3c041aa2af07be77d541538950207faada44c3cf0643c90a4e89 📄
from	0x7138a608F5CEE0db6FA98f1212c1D37706399CC2 📄
to	0xe96F6350d2059BA2E357fbDB3fa173eC18c6af16 📄
decoded output	- 📄
logs	[] 📄
raw logs	<pre>[{ "_type": "log", "address": "0x433EBAFc4A25A5377b0EA8602088AB896509dd37", "blockHash": "0x19073956bd1576d91974899cf334655e1c712bcae919e6c1328b5b6d2f619a1", "blockNumber": 10527933, "data": "0x", "index": 168,</pre>

[view on Etherscan](#)
[view on Blockscout](#)

[Verification] Contract deployed. Checking explorers for registration...

Etherscan verification skipped: API key not provided.

Please input the API key in Remix Settings - Connected Services OR Contract Verification Plugin Settings.

[Blockscout] Verification submitted. Awaiting confirmation...

[Sourcify] Verification submitted. Awaiting confirmation...

[Routescan] Verification submitted. Awaiting confirmation...

[Sourcify] Verification Successful! [View Code](#)

[Blockscout] Verification Successful! [View Code](#)

Step 8: Interact with the Contract

1. In Remix, under "Deployed Contracts," expand **LandRegistry**.
2. Run **registerLand** with ID **1**, Location **"Mumbai"**, and your address.
3. **MetaMask Action:** You must click **Confirm** in MetaMask again because this action changes the blockchain state.

non-payable **registerLand** uint256, str...

1

"Mumbai:"

500

0x7D111eCd6Ef2BB7b1D60f766eC'

Value

0

wei ▼

Gas limit

auto

0

Transact



Account 1



Transaction request

Estimated changes ?

No changes

Network

s Sepolia

Request from ?

remix.ethereum.org

Interacting with ?

0x75698...2197F

Network fee ?

0.0002 s SepoliaETH

Speed

Market ~12 sec

Cancel

Confirm

Contract interaction



Status

[View on block explorer](#)

Confirmed

[Copy transaction ID](#)

From

To

 0x7D11f...CbaE9



 0x75698...2197F

Transaction

Nonce 2

Amount **-0 SepoliaETH**

Gas Limit (Units) 118966

Gas Used (Units) 115835

Base fee (GWEI) 0.0000000028

Priority fee (GWEI) 1.5

Total gas fee 0.000174 SepoliaETH

Max fee per gas 0.000000002 SepoliaETH

Total **0.00017375 SepoliaETH**

Conclusion:-

This experiment helped in understanding how smart contracts automate processes and execute securely on the Ethereum blockchain. Using Solidity and Remix IDE, the development, deployment, and interaction with smart contracts become simple and efficient, forming the foundation for decentralized applications.